

FAIZAN POPATIYA

AI/ML Engineer | Agentic Systems | Production AI

Hyderabad, India | faizan.popatiya@cognida.ai | faizanpopatiya.com | linkedin.com/faizan-popatiya | github.com/fayzan9

SUMMARY

AI/ML engineer with 3 years of experience building and operating production AI systems, including multi-agent orchestration, retrieval pipelines, and real-time inference services. Own systems end-to-end from architecture and implementation through deployment and reliability, with production workflows serving 300+ case reviewers and reducing manual review time by 98%.

SKILLS

Software & Systems: Python, TypeScript, SQL, async/concurrent processing, WebSockets, REST API design, FastAPI, unit/integration testing, observability & monitoring, Docker, Azure Kubernetes Service (AKS), Google Cloud (Cloud Run), GitHub Actions CI/CD, PostgreSQL, pgvector, Redis

AI/ML & Agentic Systems: Multi-agent orchestration (planner-executor pattern, state management, tool execution, retries/guardrails), retrieval-augmented generation (hybrid retrieval, re-ranking, groundedness evaluation), LLM-as-judge evaluation, agent frameworks & protocols (Google ADK, MCP, function calling/tool use)

EXPERIENCE

Software Engineer, AI/ML

Mar 2023 – Present

Applied AI Software Company — Hyderabad, India

- Architected an assessment-extraction pipeline that pulls structured fields from unstructured healthcare-insurance case files using LLM-based extraction with schema validation and confidence-scored exception routing, cutting per-case review time from **8 hours to under 10 minutes** at **95% field accuracy** (audited on a 500-case sample), used by **300+ case reviewers**.
- Designed the agentic architecture for a document drafting-and-management platform: a planner agent routes requests to drafting, compliance-review, and version-control agents via typed function-calling contracts with automatic retry on tool failure, cutting first-draft turnaround from ~2 days to same-day across **10+ document types**.
- Led the engineering design of an internal conversational AI assistant used as the team's reference implementation for engineering standards — modular service boundaries, typed API contracts, and CI-gated test coverage with mandatory code review; mentored **2 engineers** through onboarding to independent shipping within 2 weeks.
- Implemented a human-feedback loop on that assistant: user corrections feed a labeled evaluation set, and LLM-as-judge scoring regression-tests response quality pre-release, cutting the incorrect-response rate ~**20%** over two iterations.
- Leading the integration of a retrieval-augmented generation pipeline into the assessment system: hybrid BM25 + dense-embedding retrieval, cross-encoder re-ranking of top-k candidates, and an LLM-judge groundedness gate that blocks ungrounded outputs pre-review — validated in a prior initiative, now in staged rollout.
- Deploy and operate platform services on **Azure Kubernetes Service (AKS)**: Dockerized builds, GitHub Actions CI/CD, request-based autoscaling, and centralized logging/metrics for incident triage, sustaining **99.5% uptime** (30-day rolling) across the assessment and document services.
- Primary technical point of contact for the client account: ran **30+ live product demos**, gathered requirements directly from claims/case-review stakeholders, and drove technical design decisions from those requirements through to shipped features.

PERSONAL PROJECTS

Multi-Agent Systems with Google ADK — github.com/fayzan9

- Implemented multi-agent workflows (Sequential/Parallel/Loop agents) with custom tool-calling functions, session-state persistence, and callback-based guardrails; authored a v1.32 migration reference that cut new-agent onboarding time from 3 days to under 4 hours for 2 internal teams.

Real-Time Voice AI Agent — Gemini Live API + Twilio — github.com/fayzan9

- Built an async, WebSocket-streamed voice agent bridging Twilio telephony audio to the Gemini Live API, handling interruption/barge-in detection, per-call state management, and backpressure-aware audio buffering; sustained **sub-800ms** round-trip latency, load-tested at **20+ concurrent calls** on autoscaled Cloud Run instances.

EDUCATION

Bachelor of Technology in Computer Science

Dr. B.R. Ambedkar Open University

Hyderabad, India